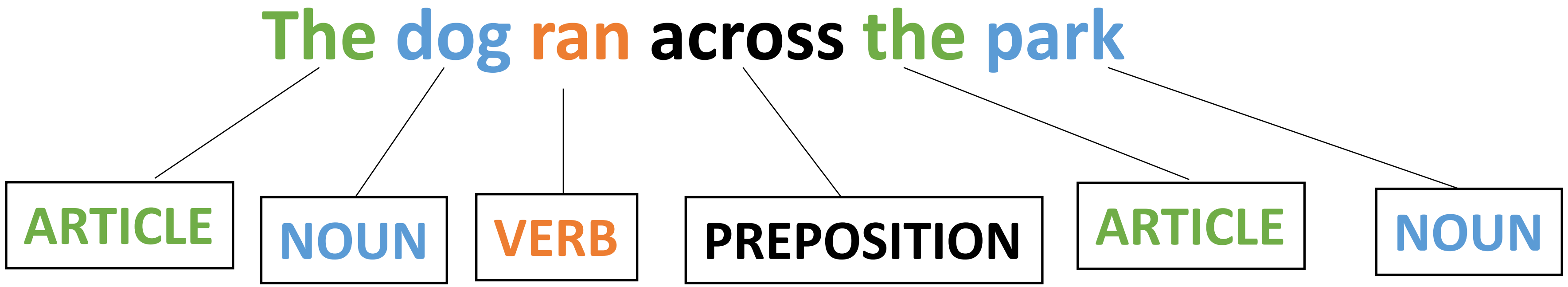
CSE110A: Compilers



- **Topic** *Review:*
  - *Naïve Scanning Method*
  - *Regular Expressions*
  - *Using regular expression's in scanners*
    - Exact match scanner
    - Start-of-string Scanner
    - Named group matcher

Review Topic:

Regular Expressions

Regular expressions

- any character ‘.’
- example using email (this is probably too general!)
- ”.*@.*\.com”

Using REs

- What if we want either the domain or user name from the email?
- We can use groups!
  - use ()s to delimitate groups
- `"(.*)@(.*\\.com)"`
- Index the resulting object with [1] and [2] to get to the user name and domain respectively

Using REs

- you can give groups id names rather than using indices
- “(?P<name>.+)@(?P<domain>.+\.com)”
- (easier to copy: "(?P<name>.+)@(?P<domain>.+\\.com)"

Review

- Why do we want REs?

Naïve Scanner

simple string stream, peek/eat model

```
class NaiveScanner:

  def token(self):
    ...
    if self.ss.peek_char() in NUMS:
      value = ""
      while self.ss.peek_char() in NUMS:
        value += self.ss.peek_char()
        self.ss.eat_char()
      return ("NUM", value)
```

```
ID     = [characters]
NUM    = [numbers]
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = [" "]
```

Shortcomings of Naïve scanner

- IDs with numbers in them?
  - `x1, y1, etc.`
  - how would you solve?

- Numbers with a decimal point in them?
  - `4.5, 9999.99998`
  - how would you solve this?

- Two character operators:
  - `++, +=`
  - how would you solve this?

Regular expressions to the rescue

Let’s write our tokens as regular expressions

- For our simple programming language

```
ID     = [characters]
NUM    = [numbers]
ASSIGN = ”=“
PLUS   = “+”
MULT   = “*”
IGNORE = [“ “, “\n”]
```

Let's write our tokens as regular expressions

- For our simple programming language

```
ID     = [characters]
NUM    = [numbers]
ASSIGN = ″=‶
PLUS   = ‶+″
MULT   = ‶*″
IGNORE = [‶ ‶, ‶\n″]
```

```
ID     = ‶[a-z][a-z0-9]*″
NUM    = ‶[0-9]+″
ASSIGN = ″=‶
PLUS   = ‶+″
PP = ‶++″
MULT   = ‶*″
IGNORE = ‶ |\n″
```

Some benefits of REs? Let's try adding some extensions:

Let’s write our tokens as regular expressions

- For our simple programming language

```
ID     = [characters]
NUM    = [numbers]
ASSIGN = ”=“
PLUS   = “+”
MULT   = “*”
IGNORE = [“ “, “\n”]
```

```
ID     = “[a-z]+”
NUM    = “[0-9]+”
ASSIGN = ”=“
PLUS   = “+”
MULT   = “*”
IGNORE = “ |\n”
```

Some benefits of REs? Let’s try adding some extensions:
** increment operator?*
** digits in IDs?*

Finishing up last lecture

- A few final thoughts:

RE examples

- **What can REs not do?**

- Nested structures, such as parathesis matching:
  - Try doing arithmetic expressions
  - You will not be able to match ()s

- Classical example: REs cannot capture same number of repeats:
  - A{N}B{N}

- REs cannot parse HTML!!!
  - One of the most upvoted answers on stackoverflow!
  - https://stackoverflow.com/questions/1732348/regex-match-open-tags-except-xhtml-self-contained-tags/1732454#1732454

How to implement an RE matcher?

- Overview: first you have to parse the RE...
  - Chicken and egg problem
  - The language of REs is not a regular language.
    - It is context free (because it has ()s)

  - But once you can parse the RE, there are several options

How to implement an RE matcher?

- parsing with derivatives
  - We discuss this in CSE211
  - Elegant solution, but difficult to make fast

- Convert to an automata
  - Learn more about this CSE103
  - A cool website
  - https://ivanzuzak.info/noam/webapps/fsm_simulator/

Review of RE Matchers to Build Scanner

- Exact match (EM) scanners
- Start-of-string (SOS) scanners
- named group (NG) scanners

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

```
ID     = “[a-z]+”
NUM    = “[0-9]+”
ASSIGN = ”=“
PLUS   = “+”
MULT   = “*”
IGNORE = “ |\n”
SEMI   = “;”
```

start with the whole string

“variable = 50 + 30 * 20;”

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

Try to match with all the tokens. No match.

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

start with the whole string

"variable = 50 + 30 * 20;"

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

Try to match with all the tokens. No match.

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

Try iteratively chopping characters from back

```
"variable = 50 + 30 * 20;"
```

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

Try to match with all the tokens. No match.

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

So on

```
"variable = 50 + 30 * 20;"
```

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

Try to match with all the tokens. No match.

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

Repeatedly delete characters from the end

```
"variable = 50 + 30 * 20;"
```

EM Scanner

- Start with the whole string, remove one character at the end until a match is found. Then return the lexeme

we can match id

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

Until we find a match at this point

```
"variable = 50 + 30 * 20;"
```

Return the lexeme

```
(ID, "variable")
```

code for exact match scanner

- Provided in your homework

EM Scanner

- Pros
  - Uses an exact RE matcher. Many RE match algorithms are exact!
- Cons
  - SLOW! Each lexeme requires many many many calls to each RE match!

SOS (Start Of String) Scanner

- We will use a new RE match function

`re.`**`fullmatch`**`(`*`pattern`*`, `*`string`*`, `*`flags=0`*`)` ¶

If the whole *string* matches the regular expression *pattern*, return a corresponding match object. Return `None` if the string does not match the pattern; note that this is different from a zero-length match.

`re.`**`match`**`(`*`pattern`*`, `*`string`*`, `*`flags=0`*`)`

If zero or more characters at the beginning of *string* match the regular expression *pattern*, return a corresponding match object. Return `None` if the string does not match the pattern; note that this is different from a zero-length match.

SOS Scanner

- The match API gives us a match starting at the beginning of the string

Feed full string into each token definition

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

```
"variable = 50 + 30 * 20;"
```

SOS Scanner

- The match API gives us a match starting at the beginning of the string

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

Feed full string into each token definition

`"variable = 50 + 30 * 20;"`

We get 1 match. We can return the lexeme

`(ID, "variable")`

SOS Scanner

- The match API gives us a match starting at the beginning of the string

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

Chop the string

`"variable = 50 + 30 * 20;"`

We get 1 match. We can return the lexeme

`(ID, "variable")`

SOS Scanner

- The match API gives us a match starting at the beginning of the string

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

Chop the string

```
" = 50 + 30 * 20;"
```

We get 1 match. We can return the lexeme

```
(ID, "variable")
```

SOS Scanner

- The match API gives us a match starting at the beginning of the string

What about the next one

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

```
" = 50 + 30 * 20;"
```

```
(ID, "variable")
```

SOS Scanner

- The match API gives us a match starting at the beginning of the string

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

What about the next one

```
" = 50 + 30 * 20;"
```

1 match: IGNORE

```
(ID, "variable")
```

SOS Scanner

- The match API gives us a match starting at the beginning of the string

```
ID     = “[a-z]+”
NUM    = “[0-9]+”
ASSIGN = ”=“
PLUS   = “+”
MULT   = “*”
IGNORE = “ |\n”
SEMI   = “;”
```

Chop the string

```
“ = 50 + 30 * 20;”
```

1 match: IGNORE

```
(ID, “variable”)
```

SOS Scanner

- The match API gives us a match starting at the beginning of the string

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

Chop the string

```
"= 50 + 30 * 20;"
```

1 match: IGNORE

```
(ID, "variable")
```

SOS Scanner

- Consideration

How to scan this string?

Try to match on each token

`“CSE110A”`

```
LETTERS = “[A-Z]+”
NUM     = “[0-9]+”
CLASS   = ”CSE110A“
```

Two matches:

```
LETTERS: “CSE”
CLASS: ”CSE110A”
```

Which one do we choose?
The longest one!

Why didn’t we have to do this for the exact match Scanner?

After each pass through token REs
we have to measure match length

SOS Scanner

- One more consideration

Within 1 RE, how does this match?

“CSE110A”

```
CLASS = “CSE|110A|CSE110A”
```

SOS Scanner

- One more consideration

Within 1 RE, how does this match?

```
"CSE110A"
```

```
CLASS = "CSE|110A|CSE110A"
```

Returns "CSE", but this isn't what we want!!!

SOS Scanner

- One more consideration

Within 1 RE, how does this match?

```
"CSE110A"
```

```
CLASS = "CSE|110A|CSE110A"
```

Returns "CSE", but this isn't what we want!!!

When using the SOS Scanner: A token definition either should not:

- *contain choices where one choice is a prefix of another*
- *order choices such that the longest choice is the first one*

SOS Scanner

- One more consideration

Within 1 RE, how does this match?

"CSE110A"

```
~~CLASS = "CSE|110A|CSE110A"~~
```

Returns "CSE", but this isn't what we want!!!

When using the SOS Scanner: A token definition either should not:

- *contain choices where one choice is a prefix of another*
- *order choices such that the longest choice is the first one*

```
CLASS = "CSE110A|110A|CSE"
```

SOS Scanner

- Pros
- Cons

SOS Scanner

- Pros
  - Much faster than EM scanner. Only 1 call to each RE per `token()` call
- Cons
  - Depends on an efficient implementation of `match()`
    - Typically provided in most RE libraries (for this exact reason)
  - Requires some care in token definitions and prefixes

SOS Scanner

We’re going to optimize this to 1 RE call!
It can really help if you have many tokens

- Pros
  - Much faster than EM scanner. Only 1 call to each RE per `token()` call
- Cons
  - Depends on an efficient implementation of `match()`
    - Typically provided in most RE libraries (for this exact reason)
  - Requires some care in token definitions and prefixes

New material for today

- *Using RE matchers to build scanners*
  - Exact match (EM) scanners
  - Start-of-string (SOS) scanners
  - **Named Group (NG) scanners**

NG (Named Group) Scanner

- We will still use the `match` API call

`re.`**`fullmatch`**`(`*`pattern`*`, `*`string`*`, `*`flags=0`*`)` ¶

If the whole *string* matches the regular expression *pattern*, return a corresponding match object. Return `None` if the string does not match the pattern; note that this is different from a zero-length match.

`re.`**`match`**`(`*`pattern`*`, `*`string`*`, `*`flags=0`*`)`

If zero or more characters at the beginning of *string* match the regular expression *pattern*, return a corresponding match object. Return `None` if the string does not match the pattern; note that this is different from a zero-length match.

NG Scanner

- Start out with token definitions
- Merge them into one RE definition

Give each group a name corresponding to its token

```
ID     = "[a-z]+"
NUM    = "[0-9]+"
ASSIGN = "="
PLUS   = "+"
MULT   = "*"
IGNORE = " |\n"
SEMI   = ";"
```

```
SINGLE_RE = "(?P<ID>[a-z]+)|
             (?P<NUM>[0-9]+)
             |(..)|"
```

NG Scanner

- to implement `token()`

Try to match the whole string to the single RE

```
SINGLE_RE = “(?P<ID>[a-z]+)|
            (?P<NUM>[0-9]+)|
            (?P<ASSIGN>=)|
            (?P<PLUS>+)|
            (?P<MULT>*)|
            (?P<IGNORE> |\n)|
            (?P<SEMI>;)”
```

```
“variable = 50 + 30 * 20;”
```

NG Scanner

- to implement `token()`

Try to match the whole string to the single RE

```
SINGLE_RE = "(?P<ID>[a-z]+)|
             (?P<NUM>[0-9]+)|
             (?P<ASSIGN>=)|
             (?P<PLUS>+)|
             (?P<MULT>*)|
             (?P<IGNORE> |\n)|
             (?P<SEMI>;)"
```

```
"variable = 50 + 30 * 20;"
```

Check the `group` dictionary in the result

NG Scanner

- to implement `token()`

Try to match the whole string to the single RE

```
SINGLE_RE = “(?P<ID>[a-z]+)|
            (?P<NUM>[0-9]+)|
            (?P<ASSIGN>=)|
            (?P<PLUS>+)|
            (?P<MULT>*)|
            (?P<IGNORE> |\n)|
            (?P<SEMI>;)”
```

```
“variable = 50 + 30 * 20;”
```

```
{”ID”      : “variable”
 “NUM”     : None
 “ASSIGN”  : None
 “PLUS”    : None
 “MULT”    : None
 “IGNORE”  : None
 “SEMI”    : None}
```

NG Scanner

- to implement `token()`

Try to match the whole string to the single RE

```
SINGLE_RE = “(?P<ID>[a-z]+)|
            (?P<NUM>[0-9]+)|
            (?P<ASSIGN>=)|
            (?P<PLUS>+)|
            (?P<MULT>*)|
            (?P<IGNORE> |\n)|
            (?P<SEMI>;)”
```

```
“variable = 50 + 30 * 20;”
```

```
{”ID”      : “variable”
 “NUM”     : None
 “ASSIGN”  : None
 “PLUS”    : None
 “MULT”    : None
 “IGNORE”  : None
 “SEMI”    : None}
```

Return the lexeme `(ID, “variable”)`

NG Scanner

- to implement `token()`

chop!

```
SINGLE_RE = "(?P<ID>[a-z]+)|
             (?P<NUM>[0-9]+)|
             (?P<ASSIGN>=)|
             (?P<PLUS>+)|
             (?P<MULT>*)|
             (?P<IGNORE> |\n)|
             (?P<SEMI>;)"
```

```
"variable = 50 + 30 * 20;"
```

```
{"ID"      : "variable"
 "NUM"     : None
 "ASSIGN"  : None
 "PLUS"    : None
 "MULT"    : None
 "IGNORE"  : None
 "SEMI"    : None}
```

Return the lexeme `(ID, "variable")`

NG Scanner

- to implement `token()`

chop!

```
SINGLE_RE = “(?P<ID>[a-z]+)|
             (?P<NUM>[0-9]+)|
             (?P<ASSIGN>=)|
             (?P<PLUS>+)|
             (?P<MULT>*)|
             (?P<IGNORE> |\n)|
             (?P<SEMI>;)”
```

```
“ = 50 + 30 * 20;”
```

How to deal with common prefixes in token definitions?

- Recall from SOS scanner:

How to scan this string?

“CSE110A”

```
LETTERS = “[A-Z]+”
NUM     = “[0-9]+”
CLASS   = ”CSE110A“
```

How to deal with common prefixes in token definitions?

- Convert to a single RE

How to scan this string?

```
"CSE110A"
```

```
SINGLE_RE = "
       (?P<LETTERS>([A-Z]+)|
       (?P<NUM>([0-9]+)|
       (?P<CLASS>CSE110A)"
```

How to deal with common prefixes in token definitions?

- Convert to a single RE

```
SINGLE_RE = “
       (?P<LETTERS>([A-Z]+)|
       (?P<NUM>([0-9]+)|
       (?P<CLASS>CSE110A)”
```

How to scan this string?

“CSE110A”

What do we think the dictionary will look like?

How to deal with common prefixes in token definitions?

- Convert to a single RE

```
SINGLE_RE = "
        (?P<LETTERS>([A-Z]+)|
        (?P<NUM>([0-9]+)|
        (?P<CLASS>CSE110A)"
```

How to scan this string?

```
"CSE110A"
```

```
{"LETTERS"  : "CSE"
 "NUM"      : None
 "CLASS"    : None
}
```

How to deal with common prefixes in token definitions?

- Convert to a single RE

```
SINGLE_RE = "
        (?P<LETTERS>([A-Z]+)|
        (?P<NUM>([0-9]+)|
        (?P<CLASS>CSE110A)"
```

```
"CSE110A"
```

```
{"LETTERS" : "CSE"
 "NUM"     : None
 "CLASS"   : None
}
```

What does this mean?

- Tokens should not contain prefixes of each other

OR

- Tokens that share a common prefix should be ordered such that the longer token comes first

How to deal with common prefixes in token definitions?

- Careful with these tokens

```
INCR = “++”
ADD  = “+”

EQ = “==“
ASSIGN = “=“
```

Ensure that you provide them in the right order so that the longer one is first!

NG Scanner

- Pros
  - FAST! Only 1 RE call per `token()`
- Cons
  - Requires a named group RE library
  - inter-token interactions need to be considered

Scanners we have discussed

- *Naïve Scanner*

- *RE based scanners*
    - Exact match (EM) scanners
    - Start-of-string (SOS) scanners
    - named group (NG) scanners

Which one to use?
Complex decision with performance, expressivity, and token requirements